

Internshala

AWS DevOps Engineer Intern Assignment

Objective

Deploy a simple website on an AWS EC2 instance and document the process.

Task 1: Create an AWS EC2 Instance (20 Marks)

Launch an Ubuntu EC2 instance, create a Security Group, allow ports 22 and 80, and connect using SSH.

Deliverables: EC2 Public IP, EC2 Dashboard screenshot, SSH login screenshot.

Task 2: Linux Basics (20 Marks)

Update packages, install Nginx, check Nginx status, restart Nginx, check disk usage, memory usage, and running processes.

Deliverables: Screenshots showing Nginx installed, running, and website accessible.

Task 3: Host a Simple Website (20 Marks)

Create a basic HTML page containing your Name, College, Branch, Email, and Current Date.

Replace the default Nginx page with it.

Deliverables: Website accessible via EC2 Public IP and browser screenshot.

Task 4: Git & GitHub (20 Marks)

Create a GitHub repository and upload the HTML file and README.md. Include EC2 setup steps, Nginx installation, and commands used.

Deliverables: GitHub repository link.

Task 5: Documentation (20 Marks)

Prepare a PDF with AWS services used, Linux commands, problems faced, learnings, and total time taken.

Bonus (+10 Marks)

Complete any ONE: Attach an Elastic IP, create another Security Group, install Docker and run hello-world, or create a shell script to restart Nginx.

Evaluation

AWS EC2 Setup:20, Linux Basics:20, Website Deployment:20, Git & GitHub:20, Documentation:20,

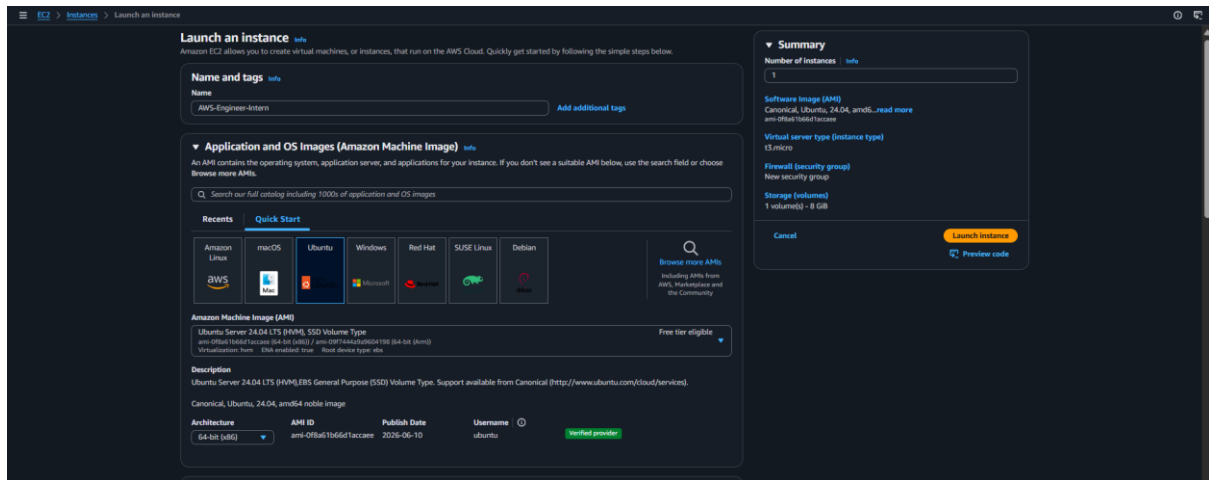
Bonus:+10. Total:100 (+10 Bonus).

Submission

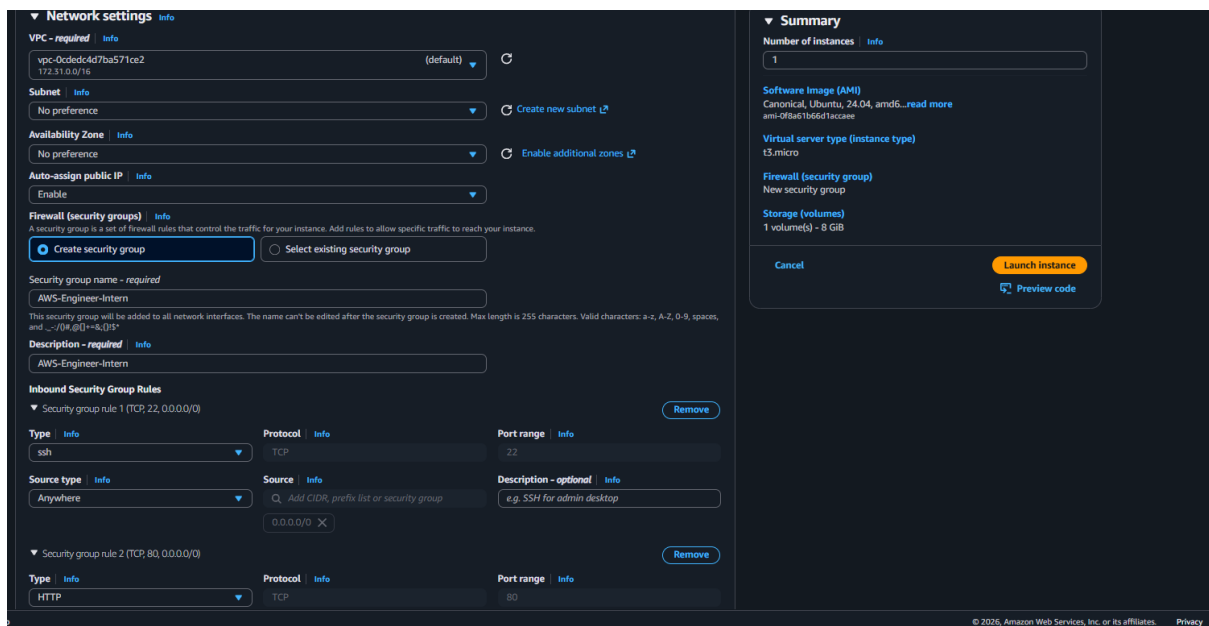
Submit GitHub repository link, PDF report, EC2 Public IP (if running), and screenshots in a ZIP file.

->create a ec2 instance

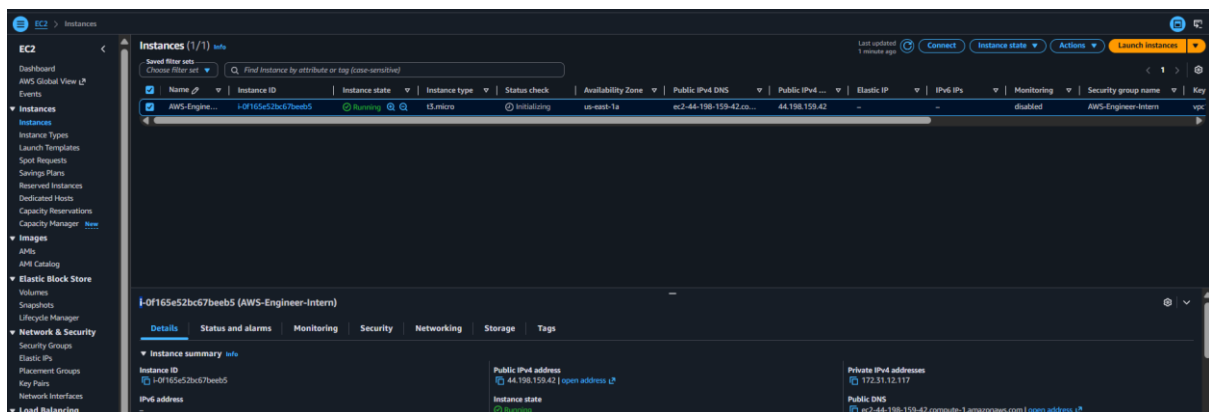
Give the name -AWS-Engineer-Intern and select the ubuntu -24.04



Create a new sg- and add -ssh 22 & -http 80



Ec2



Ssh – login

```
root@ip-172-31-12-117: ~
ED25519 key fingerprint: 1S: 5MA256:dsicsNqumPIv/h0ffEwumMhIP6Z9fMC6k12YTRtdM
This key is not known by any other names.
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes
Warning: Permanently added 'ec2-44-198-159-42.compute-1.amazonaws.com' (ED25519) to the list of known hosts.
Welcome to Ubuntu 24.04.4 LTS (GNU/Linux 6.17.0-1017-aws x86_64)

 * Documentation:  https://help.ubuntu.com
 * Management:    https://landscape.canonical.com
 * Support:       https://ubuntu.com/pro

System information as of Thu Jul 9 09:56:09 UTC 2026

System load: 0.15           Temperature: -273.1 C
Usage of /: 26.7% of 6.71GB Processes: 114
Memory usage: 25%          Users logged in: 0
Swap usage: 0%             IPv4 address for ens5: 172.31.12.117

Expanded Security Maintenance for Applications is not enabled.

0 updates can be applied immediately.

Enable ESM Apps to receive additional future security updates.
See https://ubuntu.com/esm or run: sudo pro status

The list of available updates is more than a week old.
To check for new updates run: sudo apt update

The programs included with the Ubuntu system are free software;
the exact distribution terms for each program are described in the
individual files in /usr/share/doc/*/copyright.

Ubuntu comes with ABSOLUTELY NO WARRANTY, to the extent permitted by
applicable law.

To run a command as administrator (user "root"), use "sudo <command>".
See "man sudo_root" for details.

ubuntu@ip-172-31-12-117:~$
ubuntu@ip-172-31-12-117:~$ sudo su -
root@ip-172-31-12-117:~#
```

Update packages, install Nginx, check Nginx status, restart Nginx

```
root@ip-172-31-12-117:~#
root@ip-172-31-12-117:~# sudo apt update && sudo apt upgrade -y
Hit:1 http://us-east-1.ec2.archive.ubuntu.com/ubuntu noble InRelease
Hit:2 http://us-east-1.ec2.archive.ubuntu.com/ubuntu noble-updates InRelease
Hit:3 http://us-east-1.ec2.archive.ubuntu.com/ubuntu noble-backports InRelease
Get:4 http://security.ubuntu.com/ubuntu noble-security InRelease [126 kB]
Get:5 http://security.ubuntu.com/ubuntu noble-security/main amd64 Packages [827 kB]
Get:6 http://security.ubuntu.com/ubuntu noble-security/main Translation-en [187 kB]
Get:7 http://security.ubuntu.com/ubuntu noble-security/main amd64 Components [44.9 kB]
Get:8 http://security.ubuntu.com/ubuntu noble-security/main amd64 c-n-f Metadata [11.9 kB]
Get:9 http://security.ubuntu.com/ubuntu noble-security/universe amd64 Packages [1173 kB]
Get:9 http://security.ubuntu.com/ubuntu noble-security/universe amd64 Packages [1173 kB]
Get:9 http://security.ubuntu.com/ubuntu noble-security/universe amd64 Packages [1173 kB]
Get:9 http://security.ubuntu.com/ubuntu noble-security/universe amd64 Packages [1173 kB]
Get:9 http://security.ubuntu.com/ubuntu noble-security/universe amd64 Packages [1173 kB]
Get:9 http://security.ubuntu.com/ubuntu noble-security/universe amd64 Packages [1173 kB]
Get:9 http://security.ubuntu.com/ubuntu noble-security/universe amd64 Packages [1173 kB]
Get:18 http://security.ubuntu.com/ubuntu noble-security/universe Translation-en [230 kB]
Fetched 1530 kB in 1s (3322 kB/s)
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
43 packages can be upgraded. Run 'apt list --upgradable' to see them.
[!] Killed
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
Calculating upgrade... Done
The following NEW packages will be installed:
  linux-aws-6.17-headers-6.17.0-1019 linux-aws-6.17-0-1019 linux-headers-6.17.0-1019-aws linux-image-6.17.0-1019-aws linux-modules-6.17.0-1019-aws linux-tools-6.17.0-1019-aws
The following packages will be upgraded:
  amd64-microcode ca-certificates curl gzip iproute2 kpartx libcurl3t64-gnutls libcurl4t64 libdrm-amdgpu libdrm-common libdrm2 libncurses6 libncursesw6 libnighhttp2-14 libnss3 libperl5.38t64
  libpython3.12-minimal libpython3.12-stdlib libpython3.12t64 libsqlite3-0 libtinfo6 libxml2 linux-aws linux-headers-aws linux-image-aws linux-tools-common multipath-tools ncurses-base ncurses-
  ncurses-term perl perl-base perl-modules-5.38 python3.12 python3.12-minimal tar tzdata tzdata-legacy vim vim-common vim-runtime vim-tiny xxd
43 upgraded, 6 newly installed, 0 to remove and 0 not upgraded.
38 standard LTS security updates
Need to get 114 MB of archives.
After this operation, 197 MB of additional disk space will be used.
Get:1 http://us-east-1.ec2.archive.ubuntu.com/ubuntu noble-updates/main amd64 gzip amd64 1.12-1ubuntu3.2 [99.2 kB]
Get:2 http://us-east-1.ec2.archive.ubuntu.com/ubuntu noble-updates/main amd64 ncurses-bin amd64 6.4+20240113-1ubuntu2.1 [188 kB]
Get:3 http://us-east-1.ec2.archive.ubuntu.com/ubuntu noble-updates/main amd64 libperl5.38t64 amd64 5.38.2-3.2ubuntu0.3 [4876 kB]
Get:4 http://us-east-1.ec2.archive.ubuntu.com/ubuntu noble-updates/main amd64 perl amd64 5.38.2-3.2ubuntu0.3 [231 kB]
Get:5 http://us-east-1.ec2.archive.ubuntu.com/ubuntu noble-updates/main amd64 perl-base amd64 5.38.2-3.2ubuntu0.3 [1827 kB]
Get:6 http://us-east-1.ec2.archive.ubuntu.com/ubuntu noble-updates/main amd64 perl-modules-5.38 all 5.38.2-3.2ubuntu0.3 [5110 kB]
```

Install nginx

```
root@ip-172-31-12-117:~#
root@ip-172-31-12-117:~# apt-get install nginx -y
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
The following additional packages will be installed:
  nginx-common
Suggested packages:
  fcgiwrap nginx-doc ssl-cert
The following NEW packages will be installed:
  nginx nginx-common
0 upgraded, 2 newly installed, 0 to remove and 0 not upgraded.
Need to get 568 kB of archives.
After this operation, 1599 kB of additional disk space will be used.
Get:1 http://us-east-1.ec2.archive.ubuntu.com/ubuntu noble-updates/main amd64 nginx-common all 1.24.0-2ubuntu7.1 [12.5 kB]
Get:2 http://us-east-1.ec2.archive.ubuntu.com/ubuntu noble-updates/main amd64 nginx amd64 1.24.0-2ubuntu7.13 [555 kB]
Fetched 568 kB in 0s (22.1 MB/s)
Preconfiguring packages ...
Selecting previously unselected package nginx-common.
(Reading database ... 104847 files and directories currently installed.)
Preparing to unpack .../nginx-common_1.24.0-2ubuntu7.13_all.deb ...
```

Check the status

```
root@ip-172-31-12-117:~# systemctl status nginx
● nginx.service - A high performance web server and a reverse proxy server
   Loaded: loaded (/usr/lib/systemd/system/nginx.service; enabled; preset: enabled)
   Active: active (running) since Thu 2026-07-09 10:16:18 UTC; 1min 58s ago
     Docs: man:nginx(8)
  Process: 16550 ExecStartPre=/usr/sbin/nginx -t -q -g daemon on; master_process on; (code=exited, status=0/SUCCESS)
  Process: 16552 ExecStart=/usr/sbin/nginx -g daemon on; master_process on; (code=exited, status=0/SUCCESS)
 Main PID: 16588 (nginx)
    Tasks: 3 (limit: 1013)
   Memory: 2.4M (peak: 5.3M)
      CPU: 33ms
   CGroup: /system.slice/nginx.service
           └─16588 "nginx: master process /usr/sbin/nginx -g daemon on; master_process on;"
             └─16591 "nginx: worker process"
               └─16592 "nginx: worker process"

Jul 09 10:16:18 ip-172-31-12-117 systemd[1]: Starting nginx.service - A high performance web server and a reverse proxy server...
Jul 09 10:16:18 ip-172-31-12-117 systemd[1]: Started nginx.service - A high performance web server and a reverse proxy server.
root@ip-172-31-12-117:~#
```

- Restart the nginx server and enable again

```
root@ip-172-31-12-117:~# systemctl restart nginx
root@ip-172-31-12-117:~# systemctl enable nginx
Synchronizing state of nginx.service with SysV service script with /usr/lib/systemd/systemd-sysv-install.
Executing: /usr/lib/systemd/systemd-sysv-install enable nginx
root@ip-172-31-12-117:~# systemctl status nginx
● nginx.service - A high performance web server and a reverse proxy server
   Loaded: loaded (/usr/lib/systemd/system/nginx.service; enabled; preset: enabled)
   Active: active (running) since Thu 2026-07-09 10:19:05 UTC; 21s ago
     Docs: man:nginx(8)
 Main PID: 16685 (nginx)
    Tasks: 3 (limit: 1013)
   Memory: 2.6M (peak: 2.6M)
      CPU: 16ms
   CGroup: /system.slice/nginx.service
           └─16685 "nginx: master process /usr/sbin/nginx -g daemon on; master_process on;"
             └─16686 "nginx: worker process"
               └─16687 "nginx: worker process"

Jul 09 10:19:05 ip-172-31-12-117 systemd[1]: Starting nginx.service - A high performance web server and a reverse proxy server...
Jul 09 10:19:05 ip-172-31-12-117 systemd[1]: Started nginx.service - A high performance web server and a reverse proxy server.
root@ip-172-31-12-117:~#
```

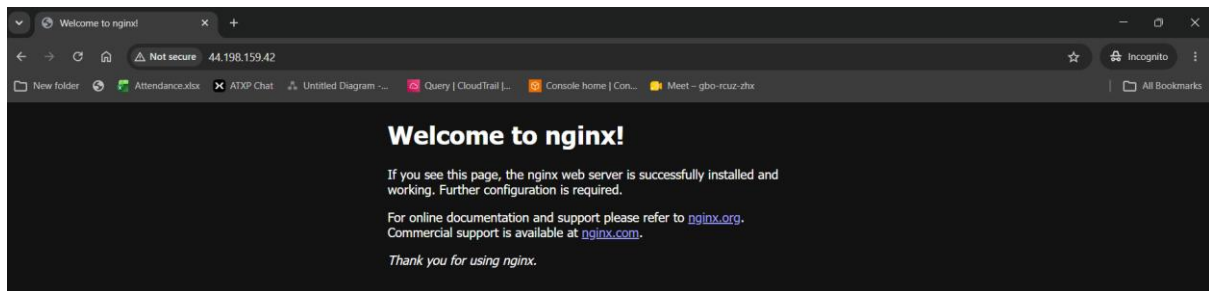
check disk usage, memory usage, and running processes

```
root@ip-172-31-12-117:~# df -h
Filesystem      Size  Used Avail Use% Mounted on
/dev/root        6.8G  2.3G  4.5G  34% /
tmpfs            456M   0  456M   0% /dev/shm
tmpfs           183M  896K  182M   1% /run
tmpfs            5.0M   0   5.0M   0% /run/lock
efivarfs        128K  3.1K  120K   3% /sys/firmware/efi/efivars
/dev/nvme0n1p6  883M  1.63M  857M  20% /boot
/dev/nvme0n1p5  105M   6.2M   99M   6% /boot/efi
tmpfs            92M   12K   92M   1% /run/user/1000

root@ip-172-31-12-117:~#
root@ip-172-31-12-117:~# free -h
              total        used        free      shared  buff/cache   available
Mem:          911Mi       416Mi       315Mi         2.7Mi       361Mi       494Mi
Swap:          0B           0B

root@ip-172-31-12-117:~# top
top - 10:26:11 up 31 min, 1 user, load average: 0.00, 0.01, 0.01
Tasks: 121 total,  1 running, 115 sleeping,  5 stopped,  0 zombie
%Cpu(s):  0.0 us,  0.2 sy,  0.0 ni, 99.8 id,  0.0 wa,  0.0 hi,  0.0 si,  0.0 st
MiB Mem :  911.3 total,  315.8 free,  416.6 used,  361.8 buff/cache
MiB Swap:   0.0 total,   0.0 free,   0.0 used,  494.7 avail Mem

  PID USER      PR  NI    VIRT    RES    SHR S  %CPU  %MEM    TIME+  COMMAND
    1 root        20   0   22548  13900   9616 S   0.0   1.5   0:03.67 systemd
    2 root        20   0     0     0     0 S   0.0   0.0   0:00.00 kthreadd
    3 root        20   0     0     0     0 S   0.0   0.0   0:00.00 pool_workqueue_release
    4 root        0 -20     0     0     0 I   0.0   0.0   0:00.00 kworker/R-rcu_gp
    5 root        0 -20     0     0     0 I   0.0   0.0   0:00.00 kworker/R-sync_wq
    6 root        0 -20     0     0     0 I   0.0   0.0   0:00.00 kworker/R-kvfree_rcu_reclaim
    7 root        0 -20     0     0     0 I   0.0   0.0   0:00.00 kworker/R-slab_flushwq
    8 root        0 -20     0     0     0 I   0.0   0.0   0:00.00 kworker/R-netns
   10 root        0 -20     0     0     0 I   0.0   0.0   0:00.00 kworker/0:0H-events_highpri
   13 root        0 -20     0     0     0 I   0.0   0.0   0:00.00 kworker/R-mm_percpu_wq
   14 root        20   0     0     0     0 S   0.0   0.0   0:00.12 ksoftirqd/0
   15 root        20   0     0     0     0 I   0.0   0.0   0:00.34 rcu_sched
   16 root        20   0     0     0     0 S   0.0   0.0   0:00.00 rcu_exp_par_gp_kthread_worker/0
   17 root        20   0     0     0     0 S   0.0   0.0   0:00.00 rcu_exp_gp_kthread_worker
   18 root        rt    0     0     0     0 S   0.0   0.0   0:00.04 migration/0
   19 root       -51   0     0     0     0 S   0.0   0.0   0:00.00 idle_inject/0
   20 root        20   0     0     0     0 S   0.0   0.0   0:00.00 cpuhp/0
   21 root        20   0     0     0     0 S   0.0   0.0   0:00.00 cpuhp/1
   22 root       -51   0     0     0     0 S   0.0   0.0   0:00.00 idle_inject/1
   23 root        rt    0     0     0     0 S   0.0   0.0   0:00.05 migration/1
   24 root        20   0     0     0     0 S   0.0   0.0   0:00.11 ksoftirqd/1
   26 root        0 -20     0     0     0 I   0.0   0.0   0:00.00 kworker/1:0H-events_highpri
   27 root        20   0     0     0     0 S   0.0   0.0   0:00.00 kdevtmpfs
   28 root        0 -20     0     0     0 I   0.0   0.0   0:00.00 kworker/R-inet_frag_wq
   29 root        20   0     0     0     0 I   0.0   0.0   0:00.00 rcu_tasks_rude_kthread
   30 root        20   0     0     0     0 I   0.0   0.0   0:00.00 rcu_tasks_trace_kthread
   31 root        20   0     0     0     0 S   0.0   0.0   0:00.00 kauditd
   32 root        20   0     0     0     0 S   0.0   0.0   0:00.00 khungtaskd
   33 root        20   0     0     0     0 I   0.0   0.0   0:00.27 kworker/u8:1-events_power_efficient
   34 root        20   0     0     0     0 S   0.0   0.0   0:00.00 oom_reaper
   35 root        0 -20     0     0     0 I   0.0   0.0   0:00.00 kworker/R-writeback
   37 root        20   0     0     0     0 S   0.0   0.0   0:00.48 kcompactd0
   38 root        25   5     0     0     0 S   0.0   0.0   0:00.00 ksmd
```



```
root@ip-172-31-12-117: /var/www/html
62 root      0 -20      0      0      0 I   0.0    0.0    0:00.00 kworker/R-er
65 root      0 -20      0      0      0 I   0.0    0.0    0:00.00 kworker/R-m1
66 root      0 -20      0      0      0 I   0.0    0.0    0:00.00 kworker/R-ip
67 root      0 -20      0      0      0 I   0.0    0.0    0:00.00 kworker/R-ks
69 root      0 -20      0      0      0 I   0.0    0.0    0:00.00 kworker/u9:0
80 root      0 -20      0      0      0 I   0.0    0.0    0:00.00 kworker/R-ch
81 root      20  0      0      0      0 S   0.0    0.0    0:00.20 jbd2/nvme0n1
[6]+  Stopped                  top
root@ip-172-31-12-117:~# cd /var/www/html
root@ip-172-31-12-117:/var/www/html# ls
index.nginx-debian.html
root@ip-172-31-12-117:/var/www/html# rm -rf index.nginx-debian.html
root@ip-172-31-12-117:/var/www/html# ls
root@ip-172-31-12-117:/var/www/html# vi index.html
root@ip-172-31-12-117:/var/www/html# |
```

```
root@ip-172-31-12-117: /var/www/html
<!DOCTYPE html>
<html>
<head>
  <title>My EC2 Website</title>
</head>
<body>

  <h1>Welcome to My EC2 Web Server</h1>

  <h2>Name: Vaibhav Mungal</h2>

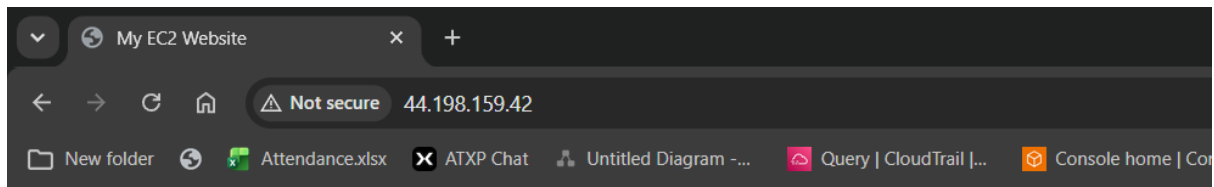
  <h3>College: MBES College of Engineering, Ambajogai</h3>

  <h3>Branch: Computer Science and Engineering</h3>

  <h3>Email: vaibhavmungal123@example.com</h3>

  <h3>Date - 09-07-2026</h3>

</body>
</html>
~
~
~
~
~
~
```



Welcome to My EC2 Web Server

Name: Vaibhav Mungal

College: MBES College of Engineering, Ambajogai

Branch: Computer Science and Engineering

Email: vaibhavmungal123@example.com

Date - 09-07-2026

Push all files on GitHub

```
root@ip-172-31-12-117:/var/www/html# git --version
git version 2.43.0
root@ip-172-31-12-117:/var/www/html# ls
index.html
root@ip-172-31-12-117:/var/www/html# git inti
git: 'inti' is not a git command. See 'git --help'.

The most similar command is
  init
root@ip-172-31-12-117:/var/www/html# git add index.html
fatal: not a git repository (or any of the parent directories): .git
root@ip-172-31-12-117:/var/www/html# git init
hint: Using 'master' as the name for the initial branch. This default branch name
hint: is subject to change. To configure the initial branch name to use in all
hint: of your new repositories, which will suppress this warning, call:
hint:
hint:   git config --global init.defaultBranch <name>
hint:
hint: Names commonly chosen instead of 'master' are 'main', 'trunk' and
hint: 'development'. The just-created branch can be renamed via this command:
hint:
hint:   git branch -m <name>
Initialized empty Git repository in /var/www/html/.git/
root@ip-172-31-12-117:/var/www/html# git add index.html
root@ip-172-31-12-117:/var/www/html# vi README.md
root@ip-172-31-12-117:/var/www/html# git add *
root@ip-172-31-12-117:/var/www/html# git commit -m "push all"
[master (root-commit) 94efa5b] push all
  Committer: root <root@ip-172-31-12-117.ec2.internal>
Your name and email address were configured automatically based
on your username and hostname. Please check that they are accurate.
You can suppress this message by setting them explicitly. Run the
following command and follow the instructions in your editor to edit
your configuration file:

    git config --global --edit

After doing this, you may fix the identity used for this commit with:

    git commit --amend --reset-author

2 files changed, 219 insertions(+)
create mode 100644 README.md
create mode 100644 index.html
root@ip-172-31-12-117:/var/www/html# git remote add origin https://github.com/Vaibhavmungal/AWS-DevOps-Engineer-Intern-Assignment.git
root@ip-172-31-12-117:/var/www/html# git remote -v
origin  https://github.com/Vaibhavmungal/AWS-DevOps-Engineer-Intern-Assignment.git (fetch)
origin  https://github.com/Vaibhavmungal/AWS-DevOps-Engineer-Intern-Assignment.git (push)
root@ip-172-31-12-117:/var/www/html# git push origin main
error: src refspec main does not match any
root@ip-172-31-12-117:/var/www/html# git branch -M main
root@ip-172-31-12-117:/var/www/html# git push origin main
Username for 'https://github.com': vaibhavmungal
Password for 'https://vaibhavmungal@github.com':
Enumerating objects: 4, done.
Counting objects: 100% (4/4), done.
Delta compression using up to 2 threads
Compressing objects: 100% (4/4), done.
Writing objects: 100% (4/4), 1.30 KiB | 1.30 MiB/s, done.
Total 4 (delta 0), reused 0 (delta 0), pack-reused 0
To https://github.com/Vaibhavmungal/AWS-DevOps-Engineer-Intern-Assignment.git
 * [new branch]      main -> main
root@ip-172-31-12-117:/var/www/html#
```